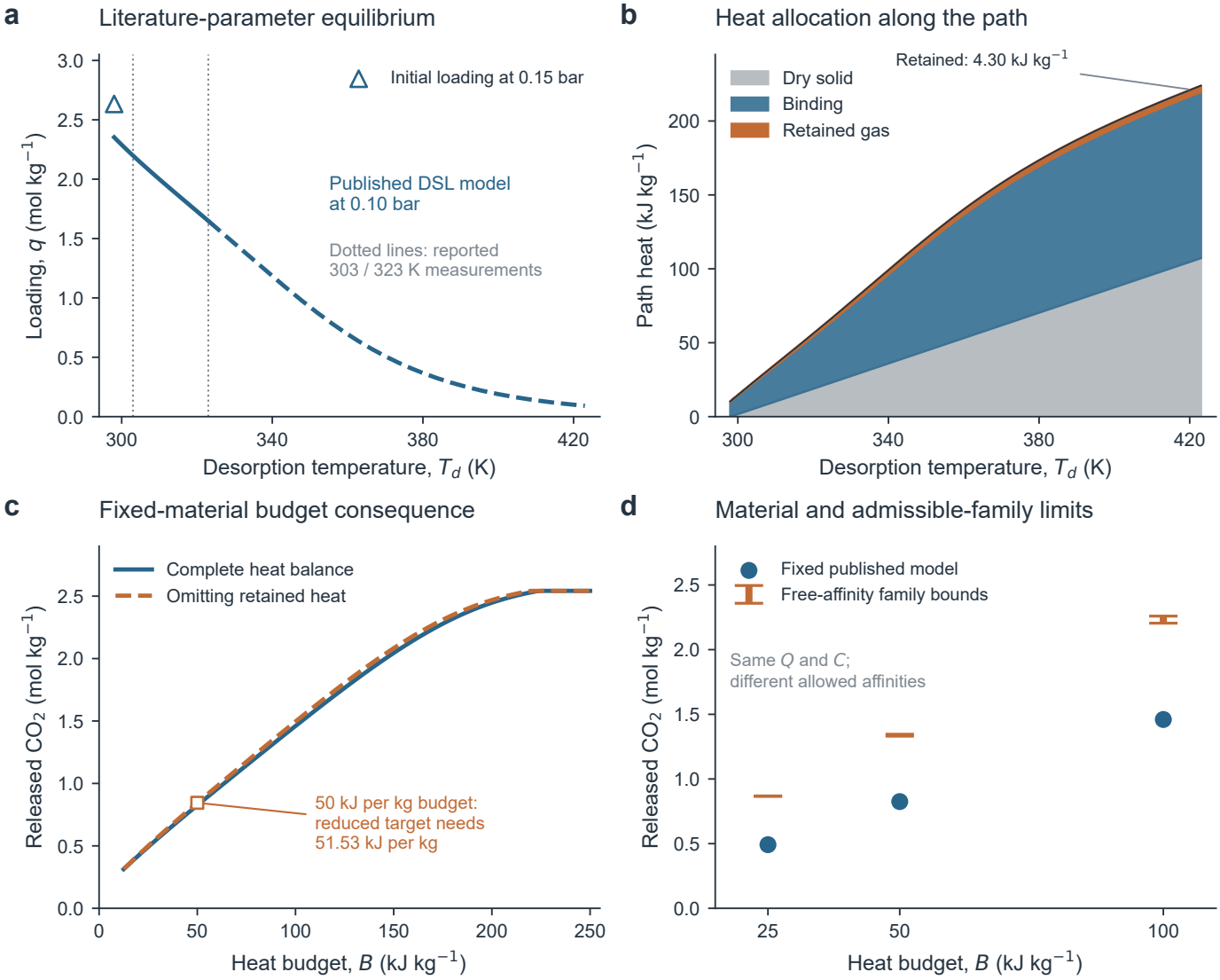

a
Literature-parameter equilibrium
Initial loading at 0.15 bar
Published DSL model
at 0.10 bar
Dotted lines: reported
303 / 323 K measurements
Loading, q (mol kg−1)
Desorption temperature, Td (K)
b
Heat allocation along the path
Retained: 4.30 kJ kg−1
Dry solid
Binding
Retained gas
Path heat (kJ kg−1)
Desorption temperature, Td (K)
c
Fixed-material budget consequence
Complete heat balance
Omitting retained heat
50 kJ per kg budget:
reduced target needs
51.53 kJ per kg
Released CO2 (mol kg−1)
Heat budget, B (kJ kg−1)
d
Material and admissible-family limits
Fixed published model
Free-affinity family bounds
Same Q and C;
different allowed affinities
Released CO2 (mol kg−1)
Heat budget, B (kJ kg−1)